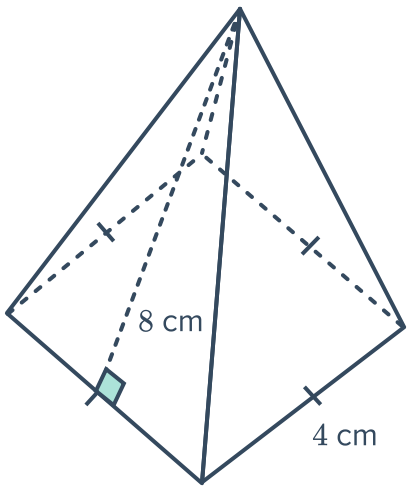
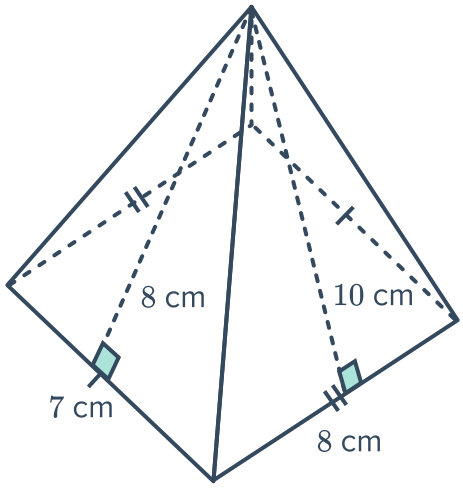
Pyramids

1 Find the surface area of the following pyramids:

a



b



2 Find the surface area of the following pyramids:

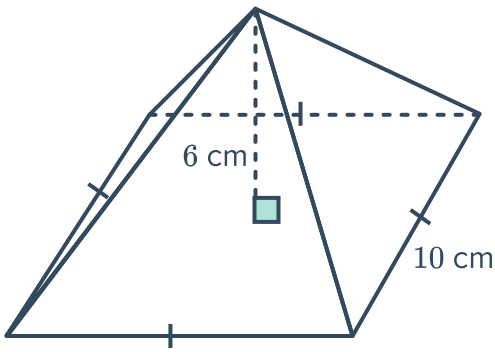
a A square pyramid with base edge 10 cm and perpendicular height 5 cm.

b A rectangular pyramid with base 10 cm by 5 cm, and perpendicular height 4 cm.

3 Consider the following square pyramid:

a Find the height of the triangular faces, correct to two decimal places.

b Find the surface area of the square pyramid correct to one decimal place.

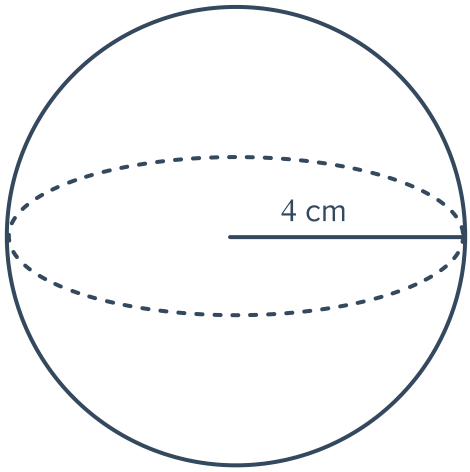


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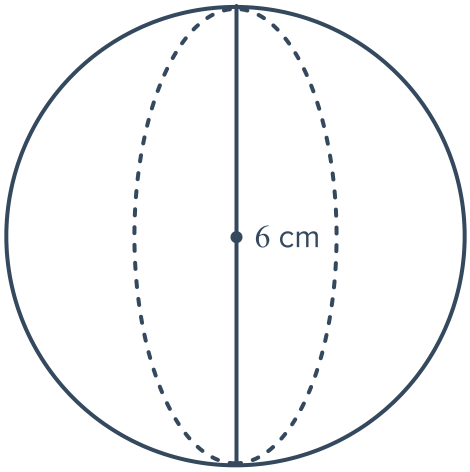
Spheres

4 Find the surface area of the following spheres, correct to two decimal places:

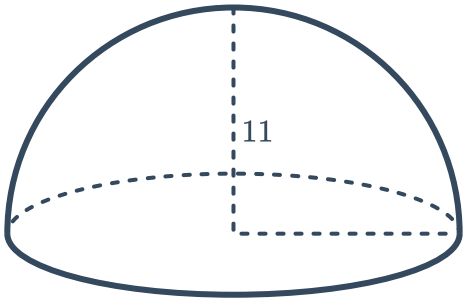
a



b



5 Find the surface area of the following hemisphere. Round your answer to three decimal places.



6 Calculate the surface area of a softball ball with a radius of $9.6\ \text{cm}$. Round your answer to two decimal places.

7 The planet Mars has a radius of $3390\ \text{km}$. Find the surface area of Mars, to the nearest whole number.

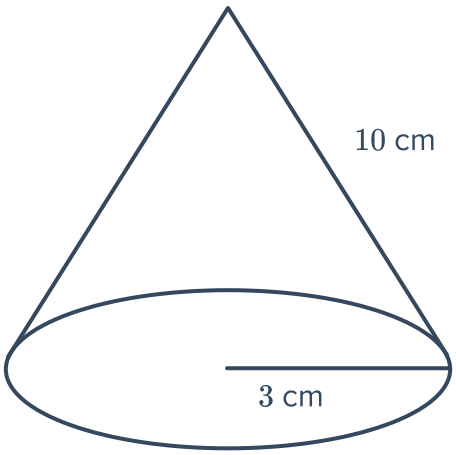
8 The planet Saturn has a radius of $58\,232\ \text{km}$, and planet Mercury has a radius of $2439.7\ \text{km}$. How many times bigger is the surface area of Saturn than Mercury? Round your answer to one decimal place.

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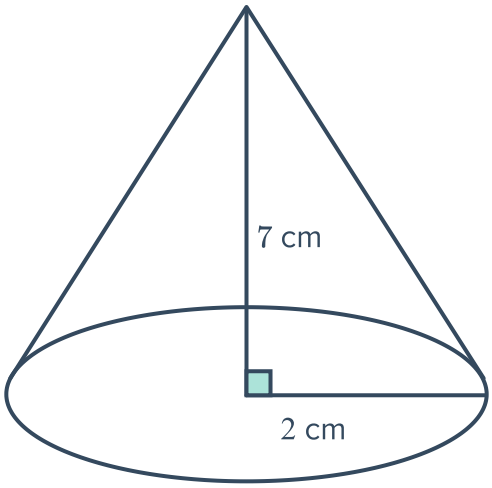
Cones

9 Find the surface area of the following cones, correct to two decimal places:

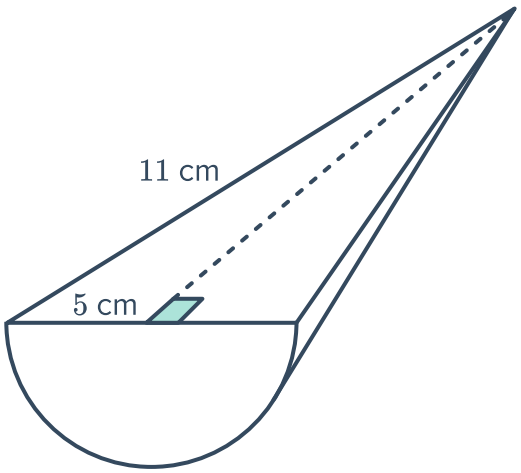
a



b



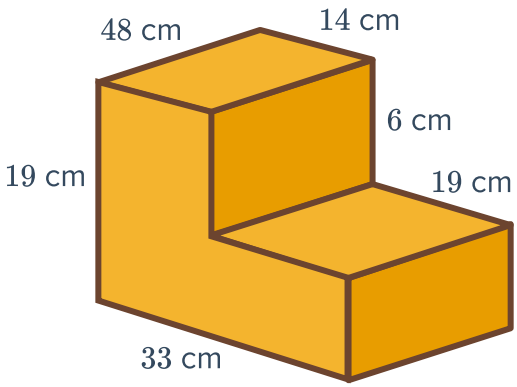
10 The diagram shows a cone of diameter 5 cm and slant height 11 cm sliced in half. Find the surface area of the solid. Round your answer to two decimal places.



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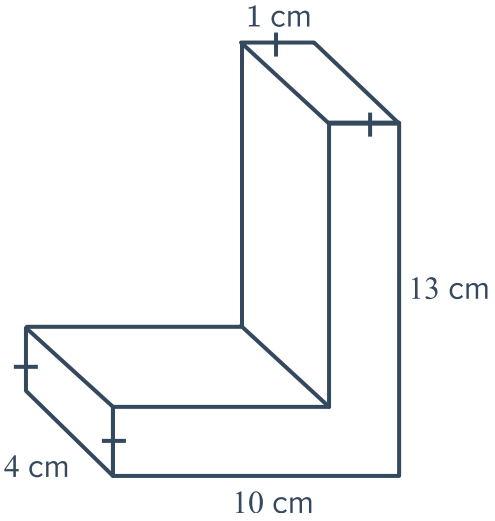
Compound solids

11 Find the surface area of the following solid:

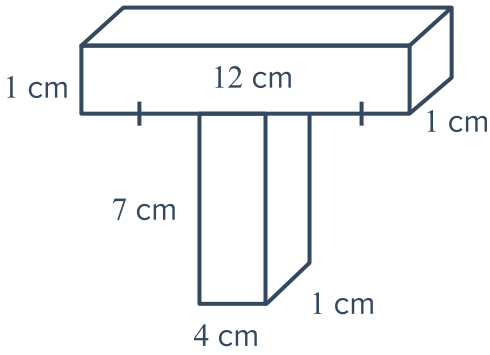


12 Find the surface area of the following solids, rounding your answers to two decimal places where necessary:

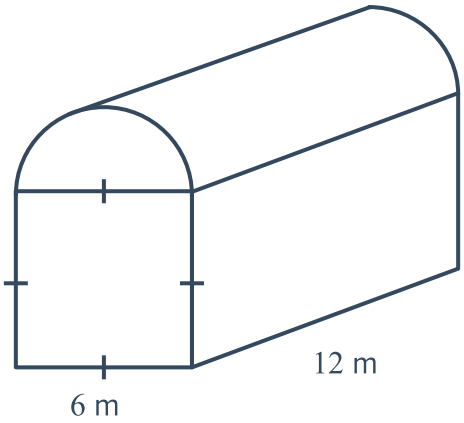
a



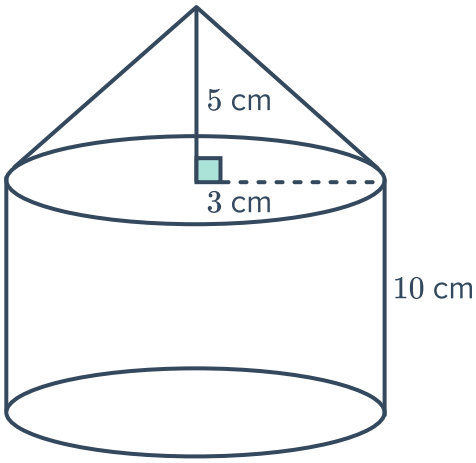
b



c

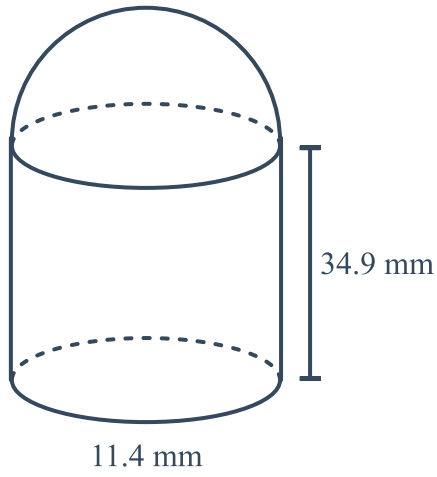


d

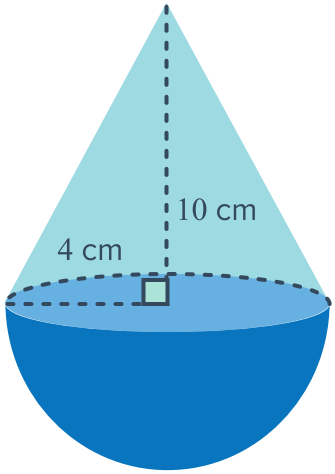


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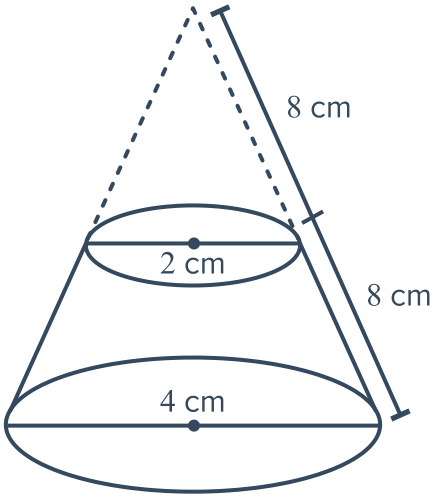
e



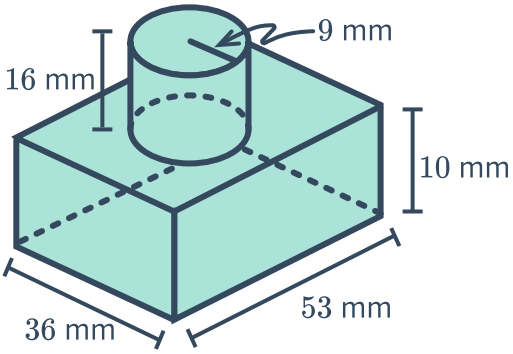
f



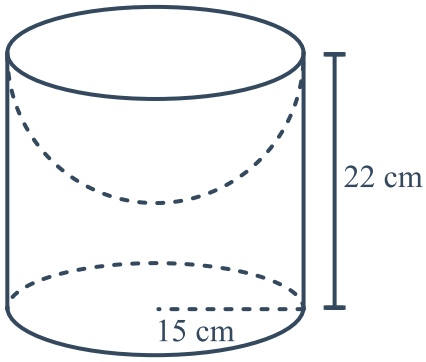
g



h



13 The following solid was constructed by removing a hemisphere from a cylinder:



Answer the following questions, rounding your answers to two decimal places.

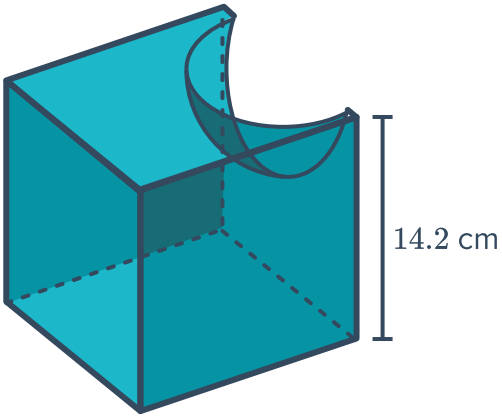
a Find the curved surface area of the hemisphere.

b Find the surface area of the open cylinder including the base.

c Hence, find the total surface area of the shape.

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14 The solid shown is constructed by cutting out a quarter of a sphere from a cube. Find its surface area if the side length is $14.2\ \text{cm}$ and the radius of the sphere is half the side length.

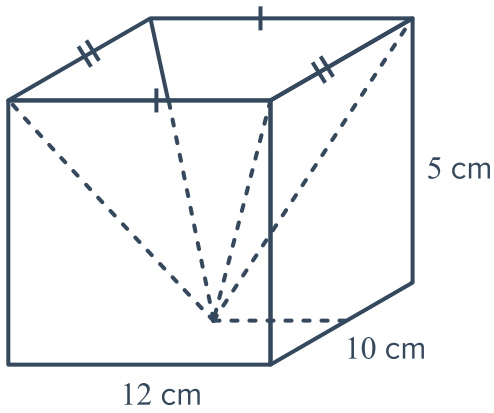


15 A pyramid has been removed from inside a rectangular prism, as shown in the figure:

a Find the perpendicular height of the triangle side with base length $12\ \text{cm}$. Round your answer to two decimal places.

b Find the perpendicular height of the triangle side with base length $10\ \text{cm}$. Round your answer to two decimal places.

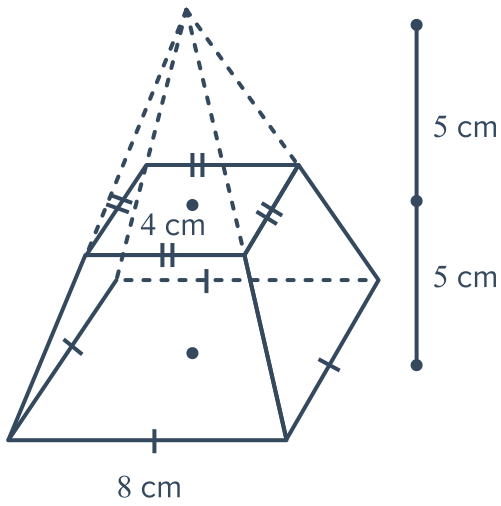
c Find the surface area of the compound solid, correct to two decimal places.



16 A small square pyramid of height $5\ \text{cm}$ was removed from the top of a large square pyramid of height $10\ \text{cm}$ leaving the solid shown:

a Find the perpendicular height of the trapezoidal sides of the new solid. Round your answer to two decimal places.

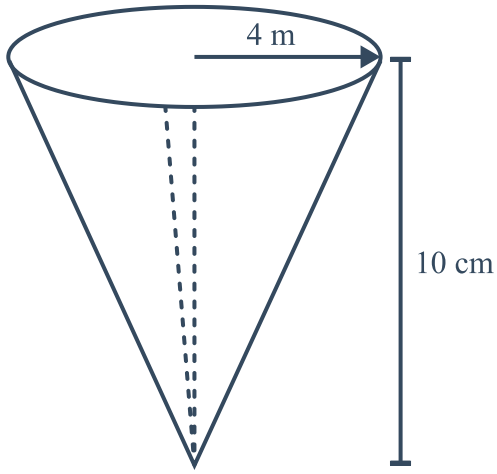
b Find the surface area of the compound solid formed, correct to one decimal place.



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Applications

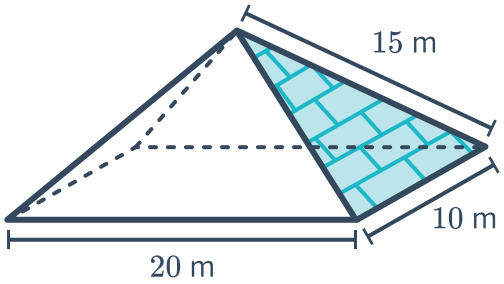
17 An ice cream cone is made by folding together a sector of pastry, with a small overlap. The dimensions of the cone are shown in the diagram:



a Find the external surface area of the cone in square centimetres. Round your answer to one decimal place.

b If the overlap adds an extra 5% to the area, how much pastry is required to produce the cone? Round your answer to the nearest square centimetre.

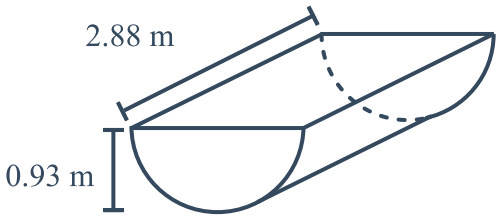
18 The roof of a large public building is in the shape of a rectangular pyramid, as shown below:



a Calculate the surface area of the roof, excluding the base of the pyramid. Round your answer to the nearest square metre.

b Each tile used on the roof has an area of $600\ \text{cm}^2$. How many tiles are used to cover the roof?

19 The diagram shows a water trough in the shape of a half cylinder:



Find the surface area of the outside of this water trough. Round your answer to two decimal places.

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20 The Louvre pyramid is a large glass and metal pyramid which serves as the entrance to the Louvre museum in Paris. It is a square pyramid with a perpendicular height of 22 m and a base length of 35 m.

If the surface of the pyramid (excluding the base) is entirely covered in glass, how many square metres of glass make up the structure? Round your answer to the nearest square metre.

21 A famouse right square pyramid is the Great Pyramid in Egypt. Its perpendicular height is approximately 139 m, its base length is 230 m, and its four triangular faces each have a height of 216 m.

Find the surface area of the Great Pyramid. Do not include the base of the pyramid in your calculation.

22 The dome of St. Paul's Cathedral is in the shape of a hemisphere, with a diameter of 31 m.

a Calculate the curved surface area of the dome. Round your answer to the nearest square metre.

b For each square metre of the internal surface area, there is approximately 43 tonnes of stone making up the dome. Calculate the total mass of the dome, to the nearest thousand tonnes.

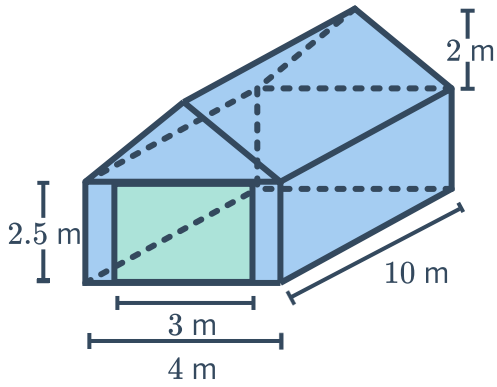
23 A steel shed is to be constructed, with dimensions as shown below. The shed is to include a rectangular cut-out at the front for the entrance.

a Determine the surface area of the shed. Round your answer to one decimal place.

b Construction of the shed requires an additional 0.1 m^2 of sheet metal for each 1 m^2 of surface area, due to overlaps and wastage.

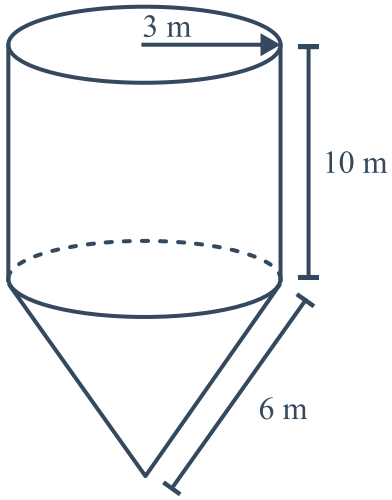
How much sheet metal is required to construct this shed? Round your answer up to the nearest square metre.

c If the steel sheets cost $\$18$ per square metre, calculate the total cost of the steel required to build this shed.



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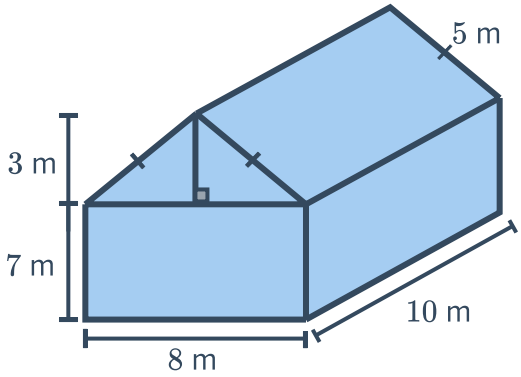
24 A grain silo has the shape of a cylinder attached to a cone, with dimensions as shown in the diagram on the right:



a Find the surface area of the silo, to the nearest square metre, assuming that the top is closed.

b The silo is manufactured out of sheet metal that has a mass of 2.4 kg/m ^{2}. Find the total mass of the silo to the nearest kilogram.

25 The given diagram shows the design for a marquee (tent). The roof of the marquee has a height of 3 m. The material for the marquee costs $\$44/\text{m}^2$.



a Find the total surface area of the marquee. Do NOT include the floor.

b Find the total cost of the marquee material.

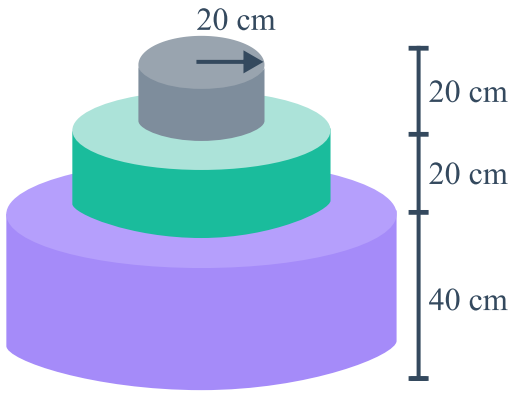
26 Xavier has been hired to wallpaper the walls and ceiling of a living room. The room is 9.7 m long, 4.2 m wide, and 2.91 m high. There is one window that is 2.3 m by 1.1 m, two windows that are 0.97 m by 1.1 m, and two doors that are 1.34 m by 2.6 m.

Xavier needs to wallpaper everything except the floor, the windows, and the doors. Find the total surface area that Xavier needs to wallpaper, rounding your answer to two decimal places.

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27 A wedding cake consists of three cylinders stacked on top of each other. The dimensions are as follows:

- The top layer has a radius and height of 20 cm.
- The middle layer has the same height as the top layer and a radius that is double that of the top layer.
- The bottom layer has a height that is double that of the top layer, and a radius that is triple that of the top layer.

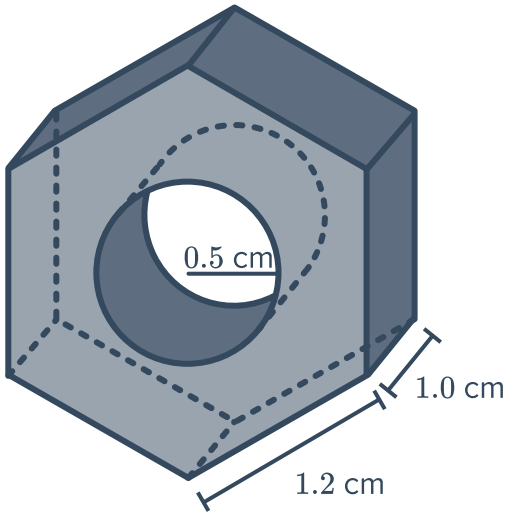


All the sides and top surfaces are to be covered in icing, but not the base. Find the surface area of the cake that needs to be iced. Round your answer to the nearest square centimetre.

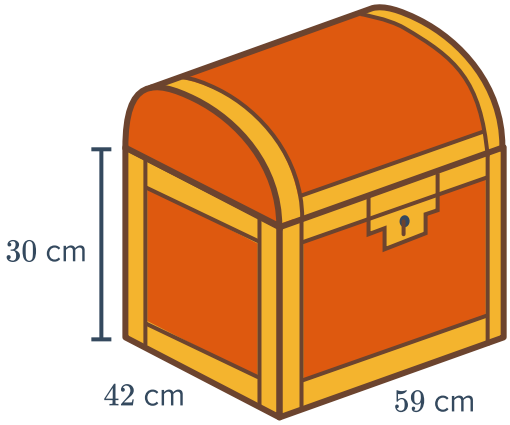
28 A company manufactures nuts shaped like regular hexagonal prisms, with cylindrical bolt holes cut out of the centre, as shown below:

a The total surface area of a nut before the bolt hole is drilled is 14.7 cm^2. Find the surface area of a single nut after the bolt hole is drilled out, including the inside surface area of the hole. Round your answer to one decimal place.

b Each nut that is manufactured requires a zinc coating to prevent corrosion. If 1 kg of zinc is enough to coat a surface area of 1 m^2, how many nuts can be coated with 1 kg of zinc? Round your answer to the nearest whole number.



29 The lid of this treasure chest is found to be exactly one half of a cylindrical barrel. Find the surface area of the chest, correct to two decimal places.



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